

Analysis group meeting

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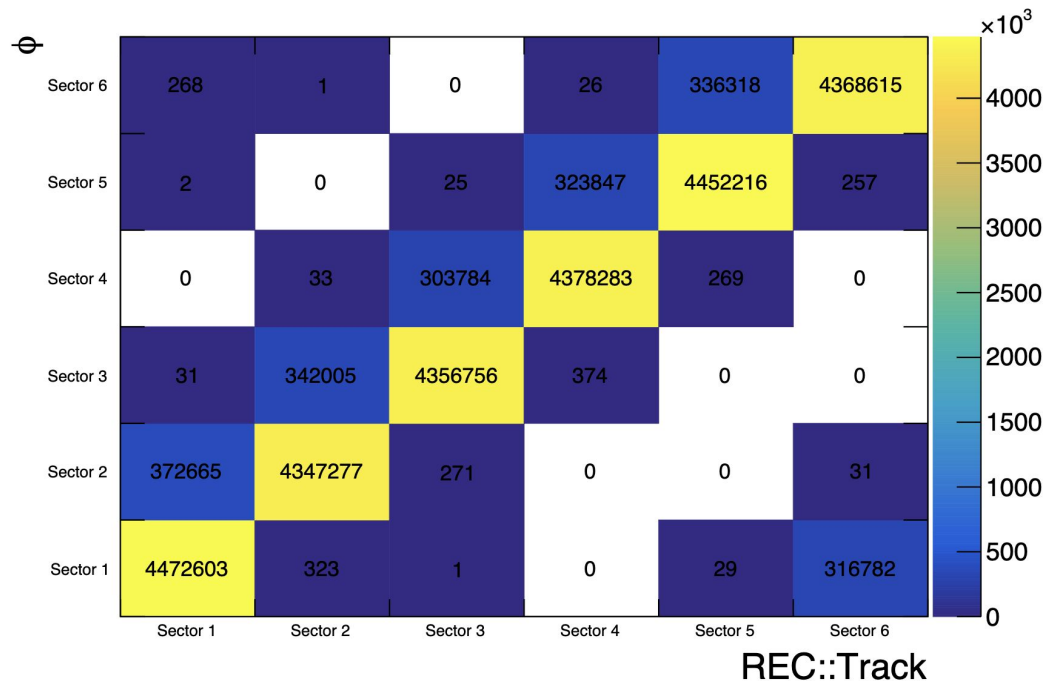
Vertex study: ϕ vs REC::Track

2

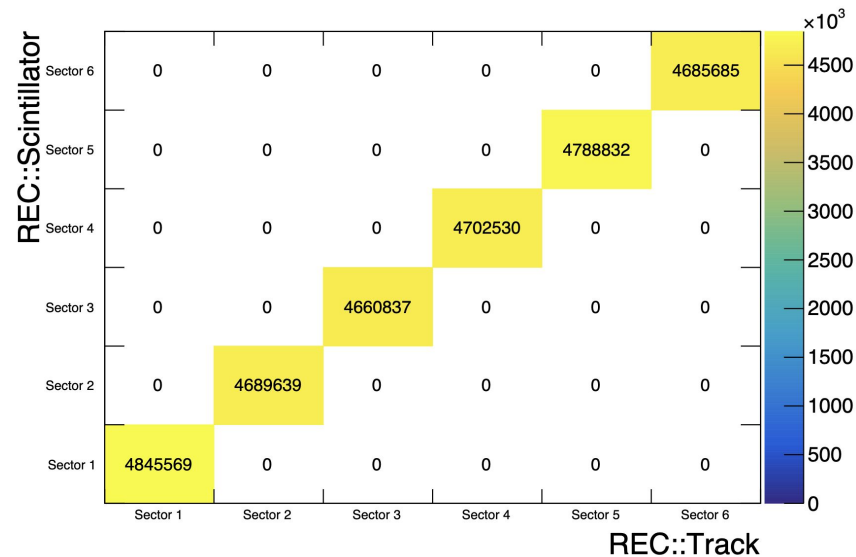
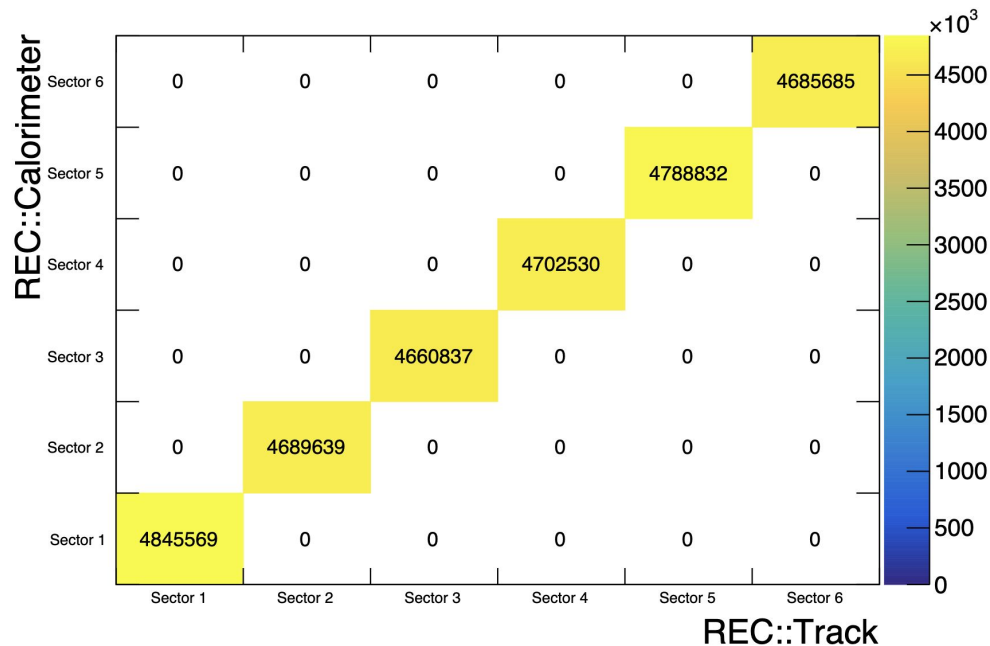
Sector information can come from the REC::Track bank or the ϕ information for the electron. Use Pass0v7 data, CuSn target.

Some mistake made last time:
sector information is from REC::Track bank
In the fiducial cuts of RG-A, they use the x
and y positions in the DC.

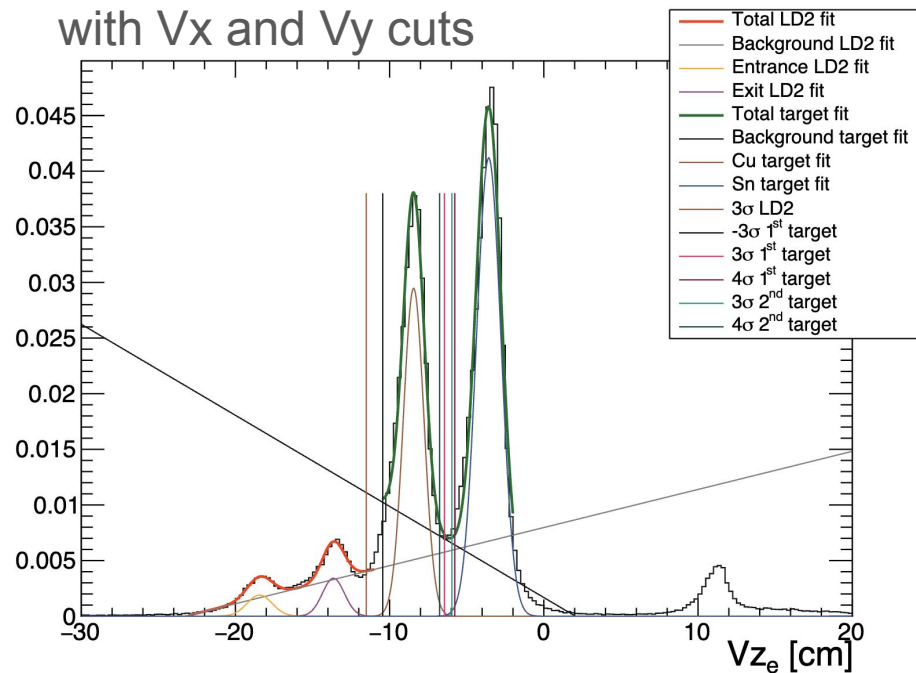
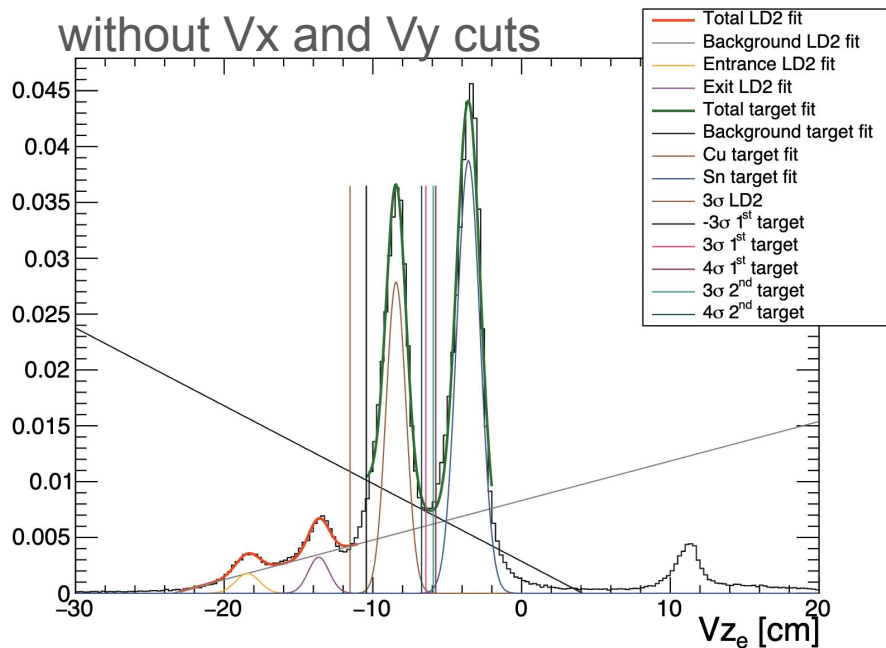
I haven't found how they get the sector
information in some analysis [1,2,3](#).
I have discussed this with Raffaella, and she
recommends using the information in the
REC::Track, REC::Calorimeter, or
REC::Scintillator.



Vertex study: REC::Scintillator and REC::Calorimeter



Vertex study: V_z with and without V_x and V_y cuts



Use Pass0v7 data, run 018561, 018563, and 018564, CuSn target

without Vx and Vy cuts:

μ_1 :-8.44092 cm, μ_2 :-3.57345 cm
 σ_1 :0.666788 cm, σ_2 :0.787793 cm

with Vx and Vy cuts:

μ_1 :-8.4429 cm, μ_2 :-3.5734 cm
 σ_1 :0.666602 cm, σ_2 :0.796239 cm

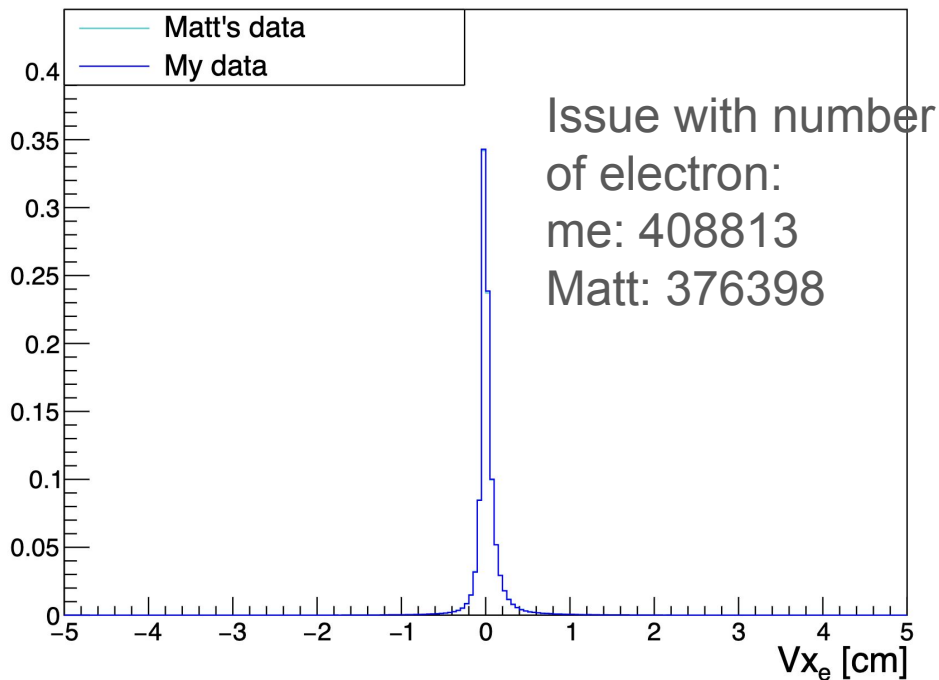
Small difference between, can remove at least for the electron, the Vx and Vy cuts.
Check on the charged pions.

Vertex study: Comparison Matt and me

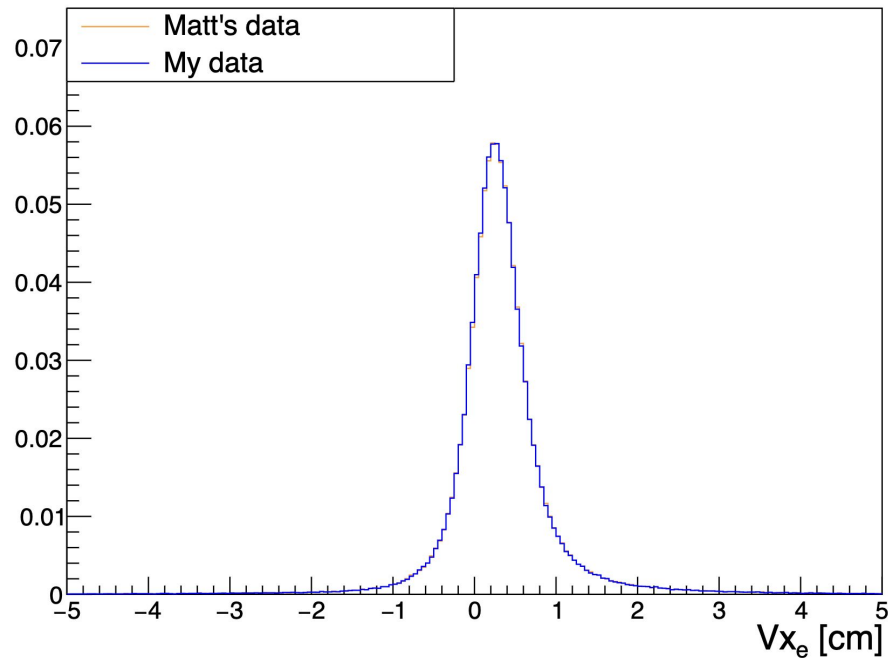
6

Use Pass0v7 data, run 018561, 018563, and 018564, CuSn target

Electron, sector 1

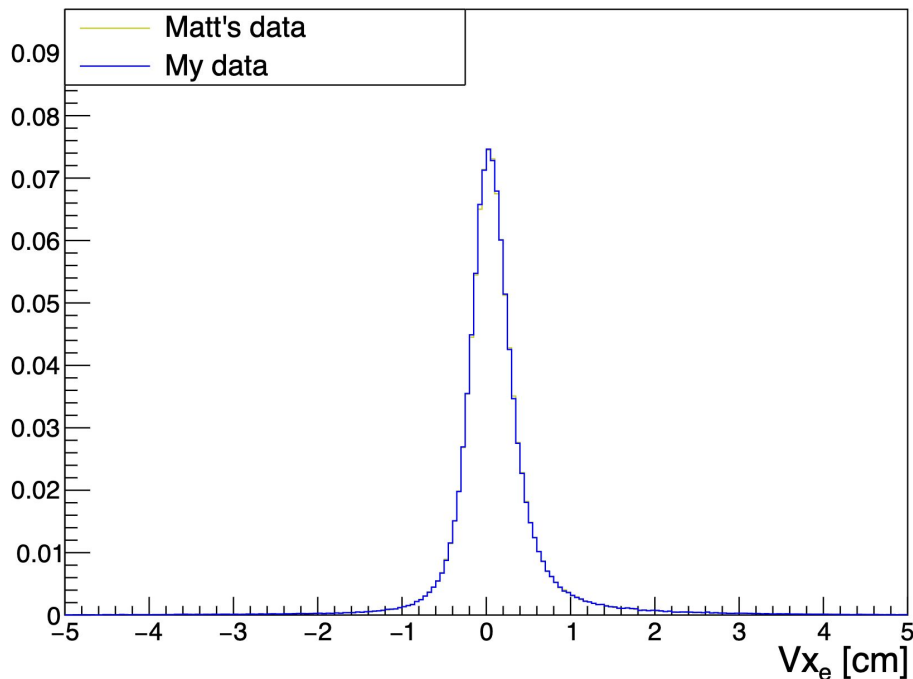


Electron, sector 2

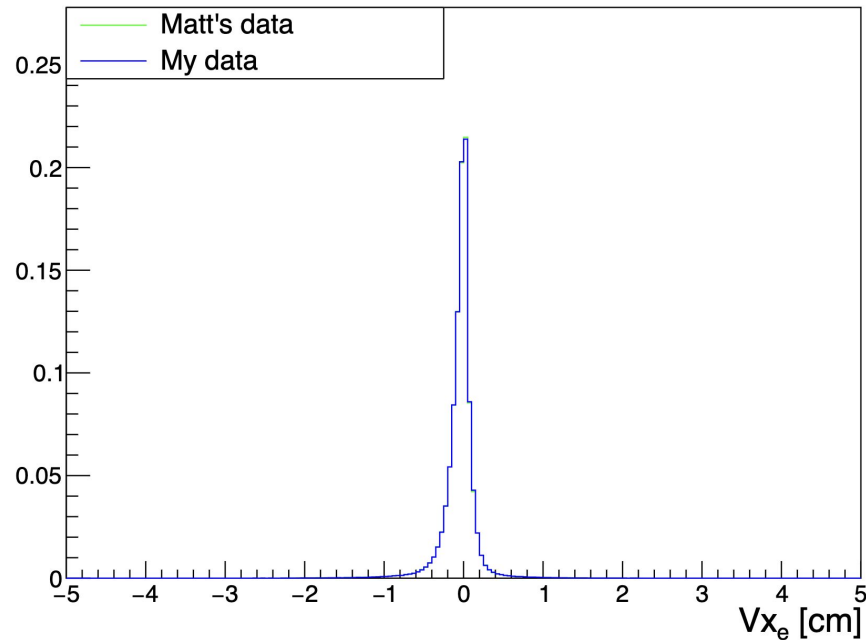


Use Pass0v7 data, run 018561, 018563, and 018564, CuSn target

Electron, sector 3



Electron, sector 4

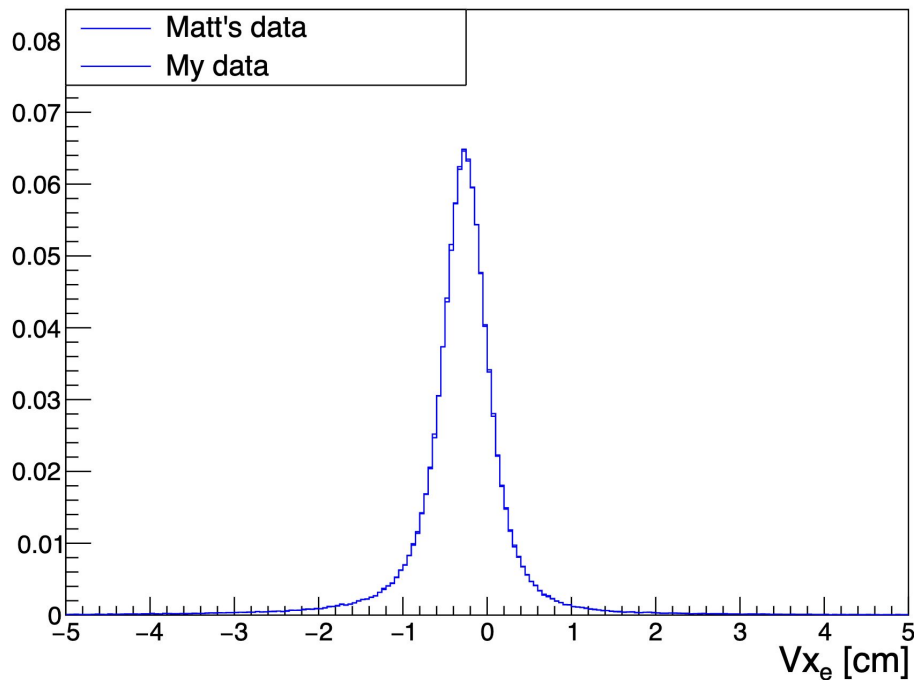


Vertex study: Comparison Matt and me

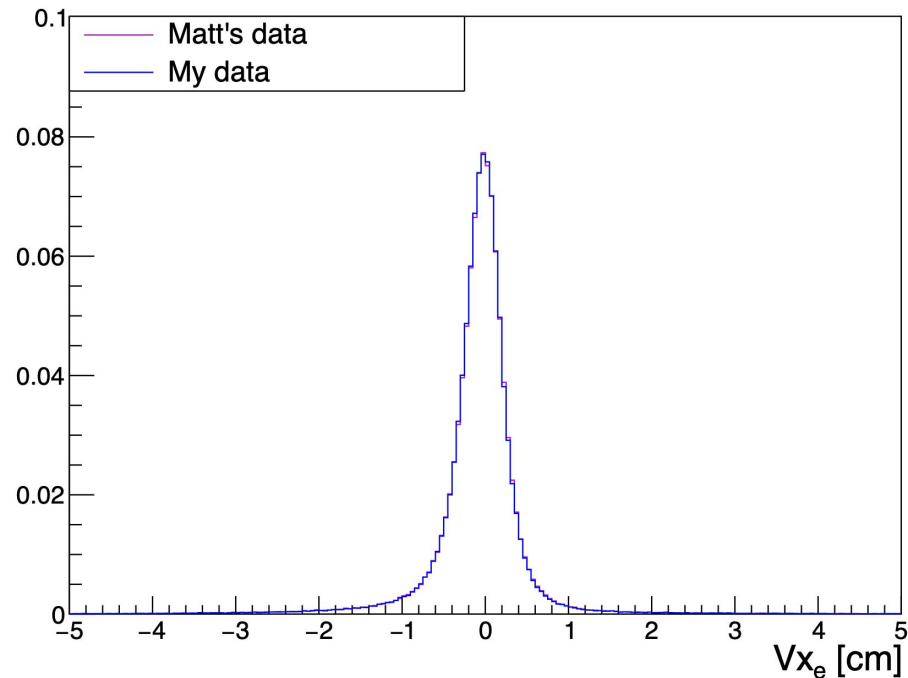
8

Use Pass0v7 data, run 018561, 018563, and 018564, CuSn target

Electron, sector 5

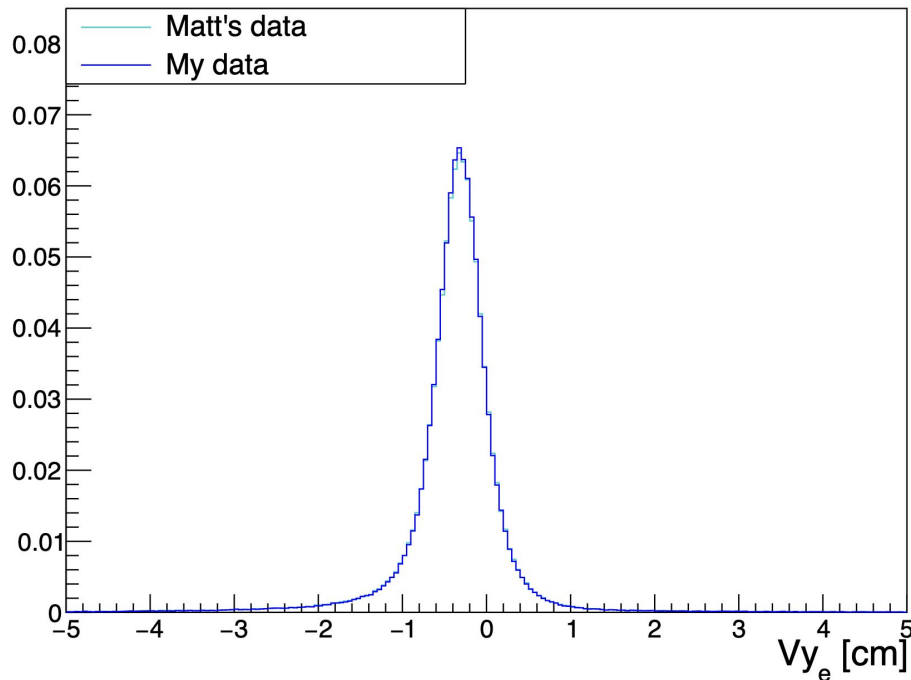


Electron, sector 6

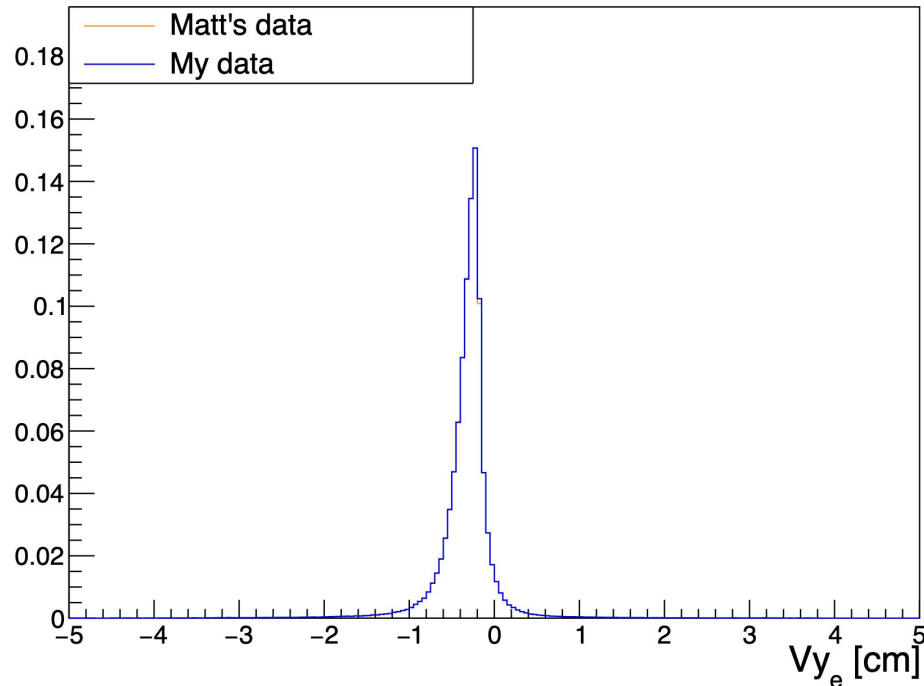


Use Pass0v7 data, run 018561, 018563, and 018564, CuSn target

Electron, sector 1

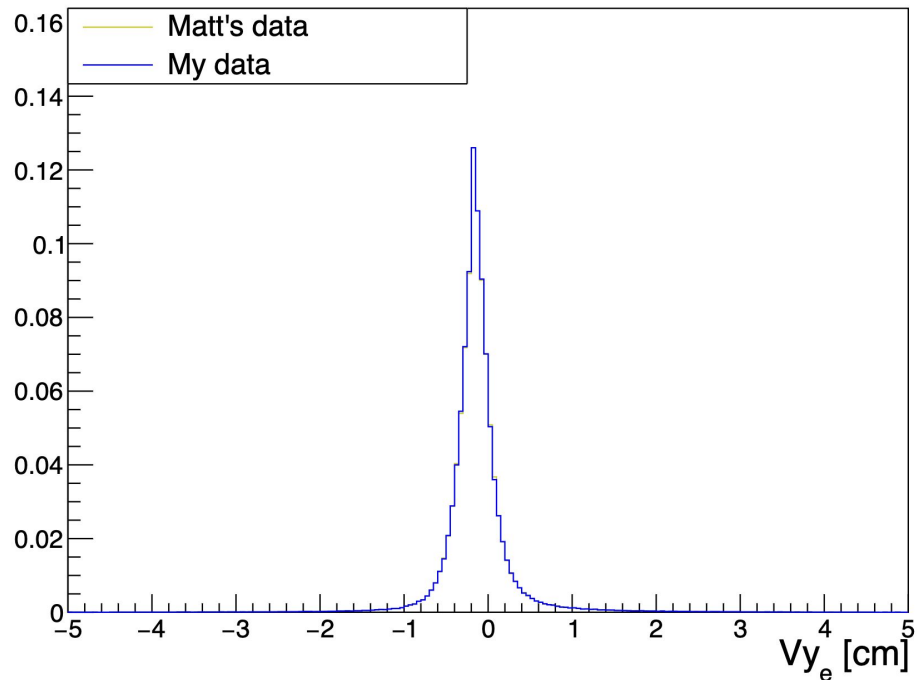


Electron, sector 2

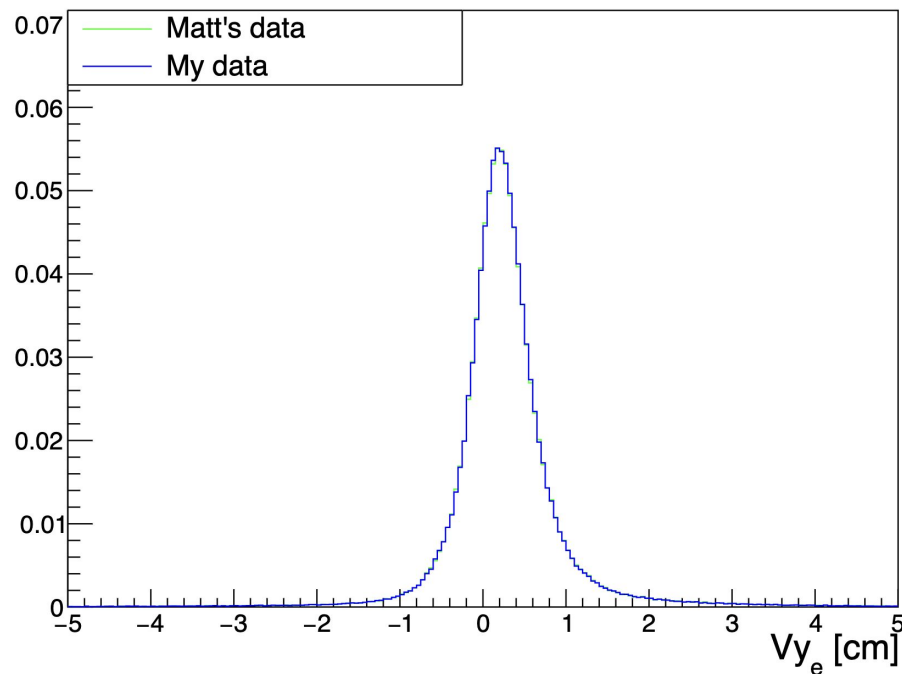


Use Pass0v7 data, run 018561, 018563, and 018564, CuSn target

Electron, sector 3

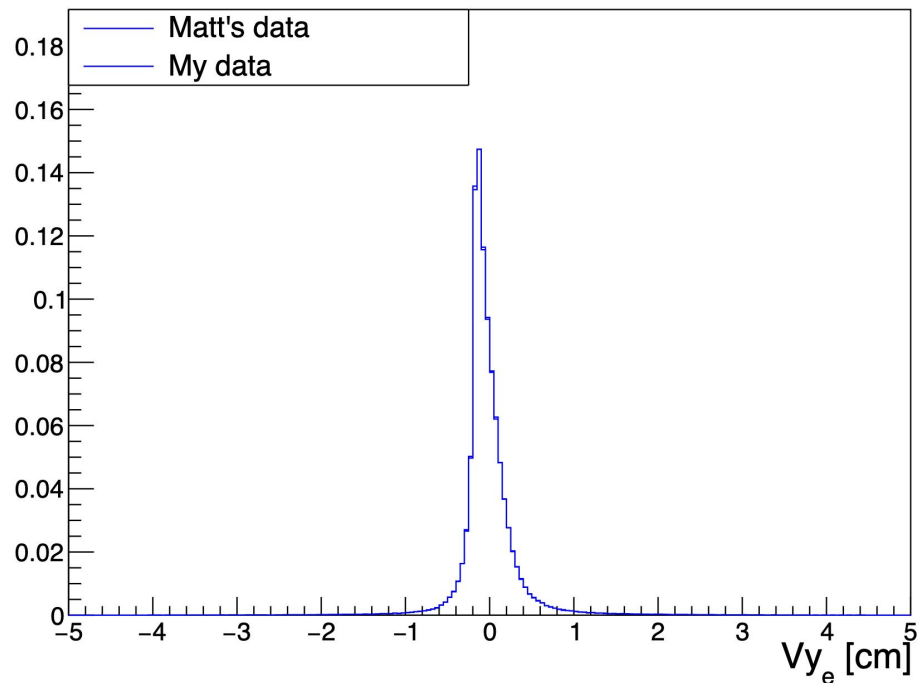


Electron, sector 4

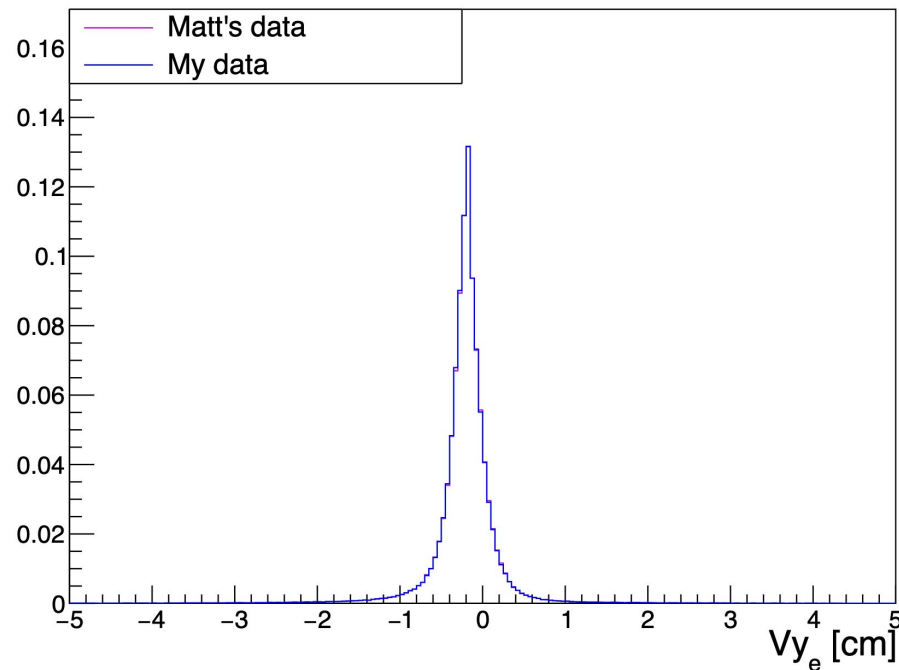


Use Pass0v7 data, run 018561, 018563, and 018564, CuSn target

Electron, sector 5

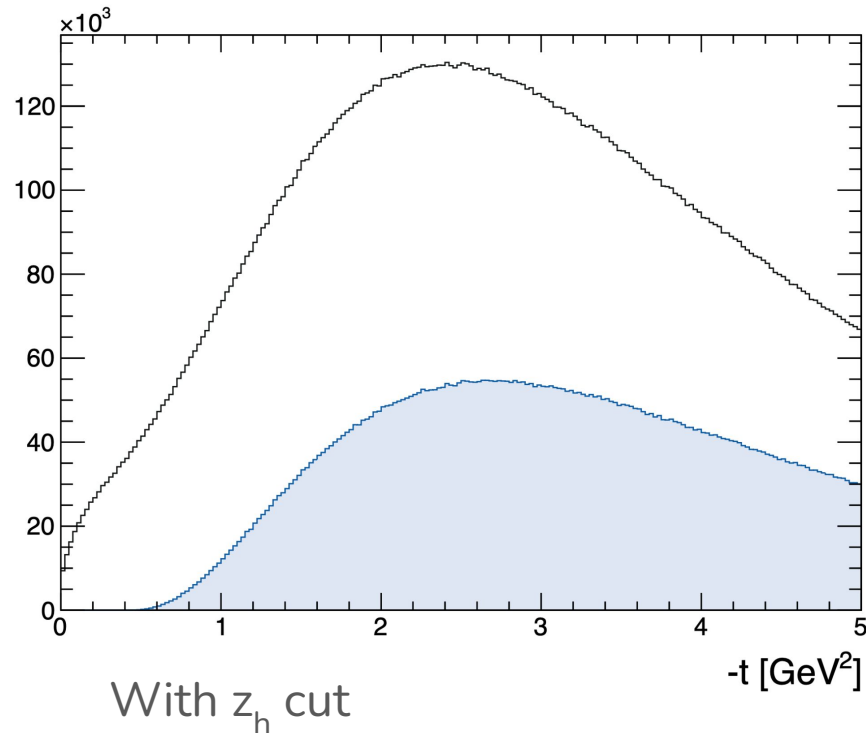


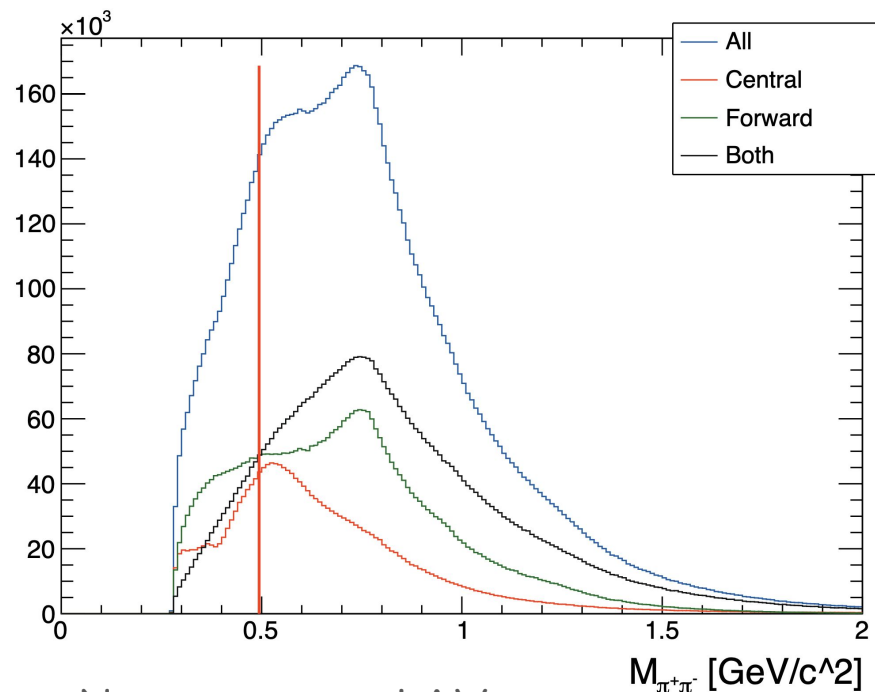
Electron, sector 6



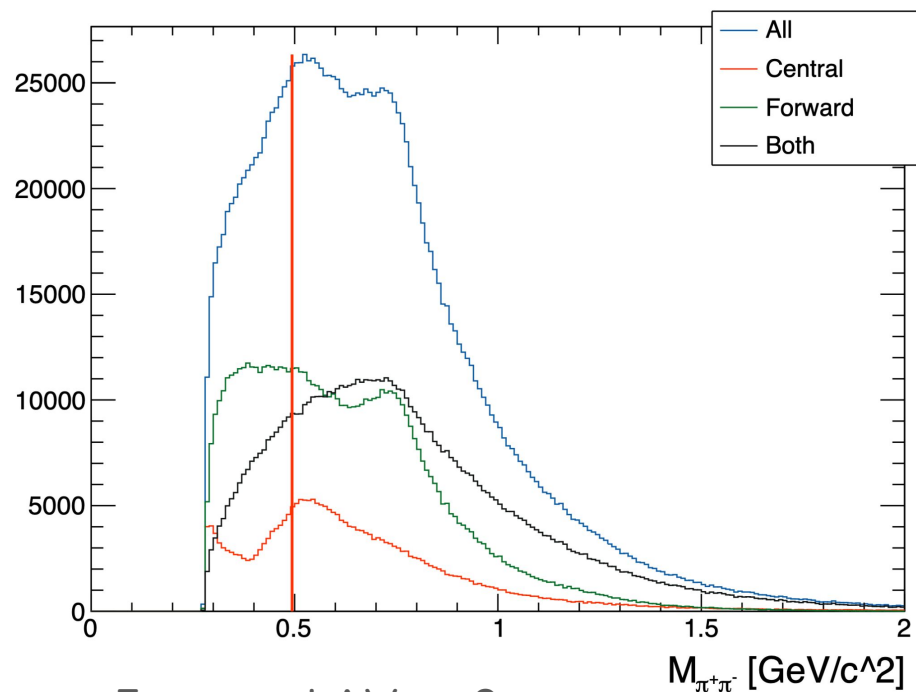
Data: Pass0v7, LD2 target

- Electron cuts:
 - $\text{pid} = 11$ and $\text{status} < 0$
 - $-5 < \chi^2_{\text{pid}} < 5$
 - $-10 < v_z < 0$ cm
- Pions cuts:
 - $\text{pid} = 211, -211$
 - $-10 < \chi^2_{\text{pid}} < 10$
 - $-10 < v_z < 0$ cm
- Kinematical cuts:
 - $Q^2 > 1 \text{ GeV}^2$ to select the DIS regime
 - $W > 2 \text{ GeV}$ to avoid the resonance region
 - $y < 0.85$ to avoid large contribution from radiative processes
 - $z_h < 0.9$ to reduce diffractive p^0
 - $-t > 0.4 \text{ GeV}^2$ to reduce diffractive p^0



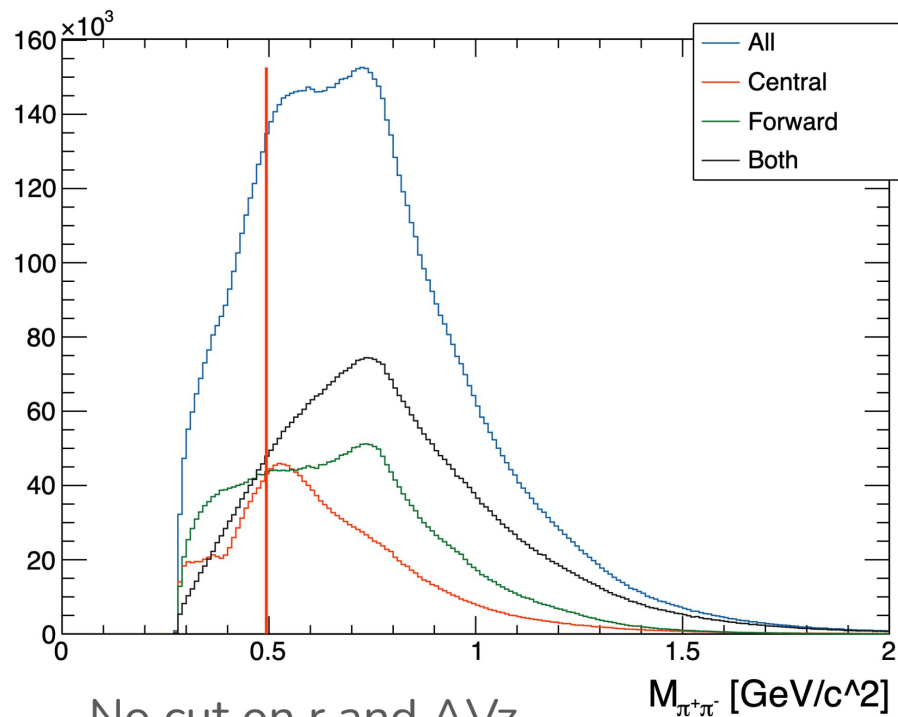


No cut on r and ΔVz

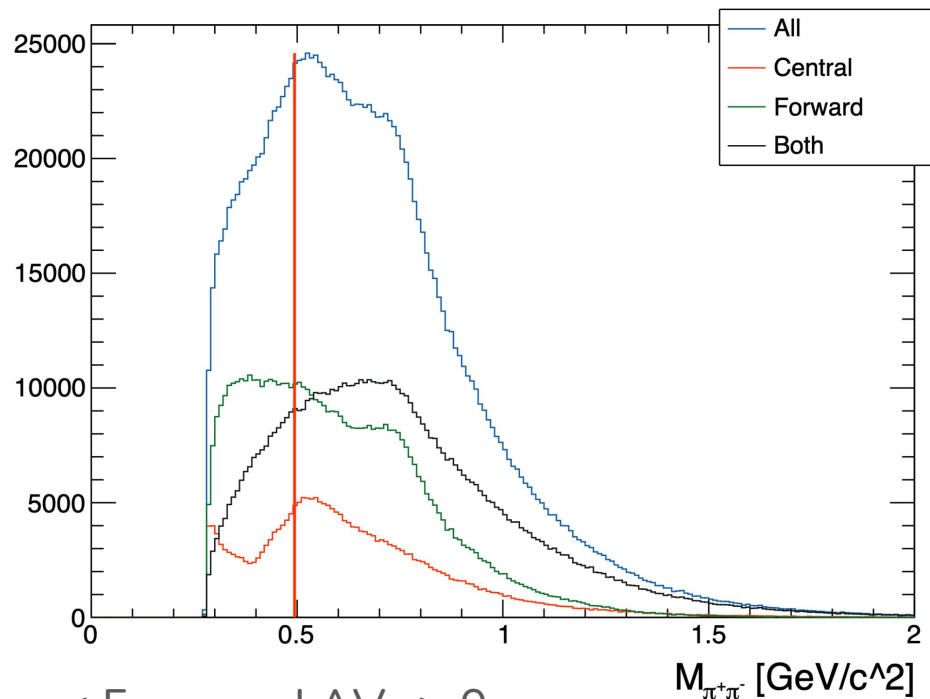


$r < 5\text{mm}$ and $\Delta Vz > 0\text{cm}$

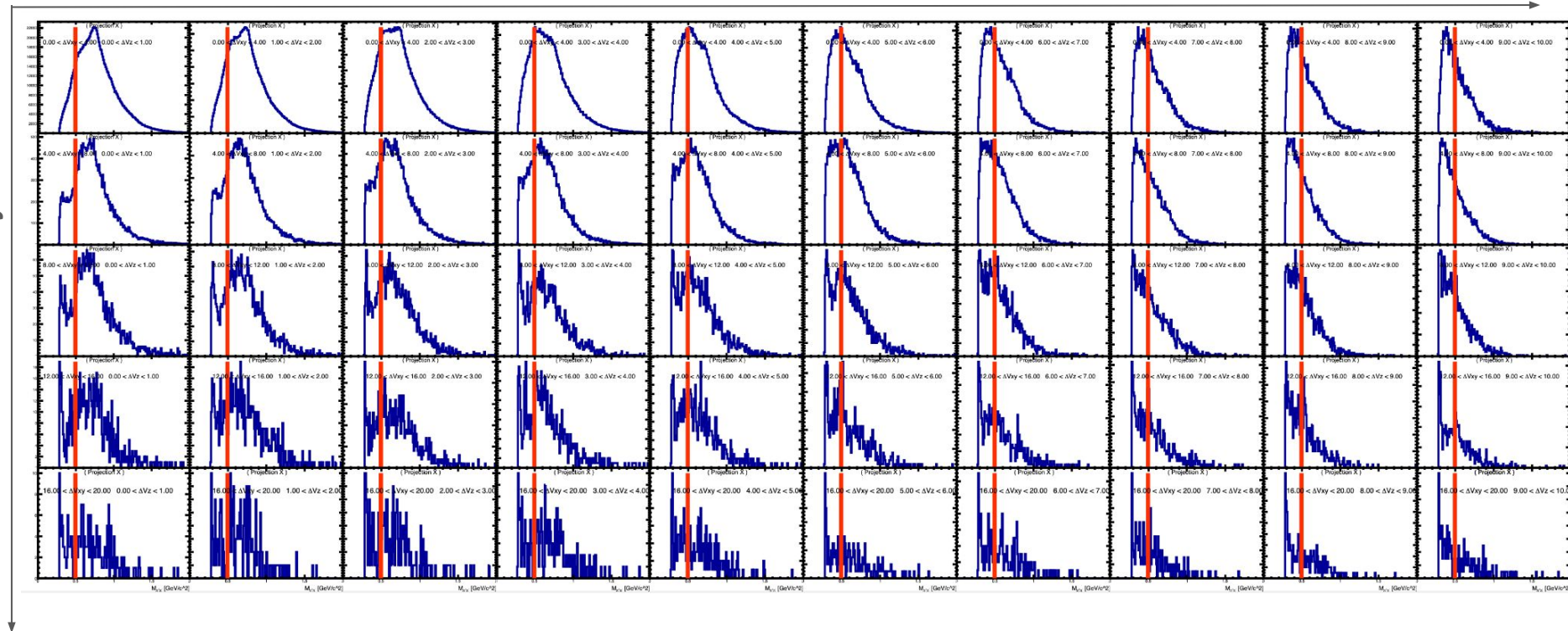
RG-D study: K^0 analysis, with cuts on z_h and $-t$



No cut on r and ΔVz
 Small reduction of the ρ^0 peak



$r < 5\text{mm}$ and $\Delta Vz > 0\text{ cm}$

ΔV_z
 ΔV_{xy}


RG-D study: π^0 analysis

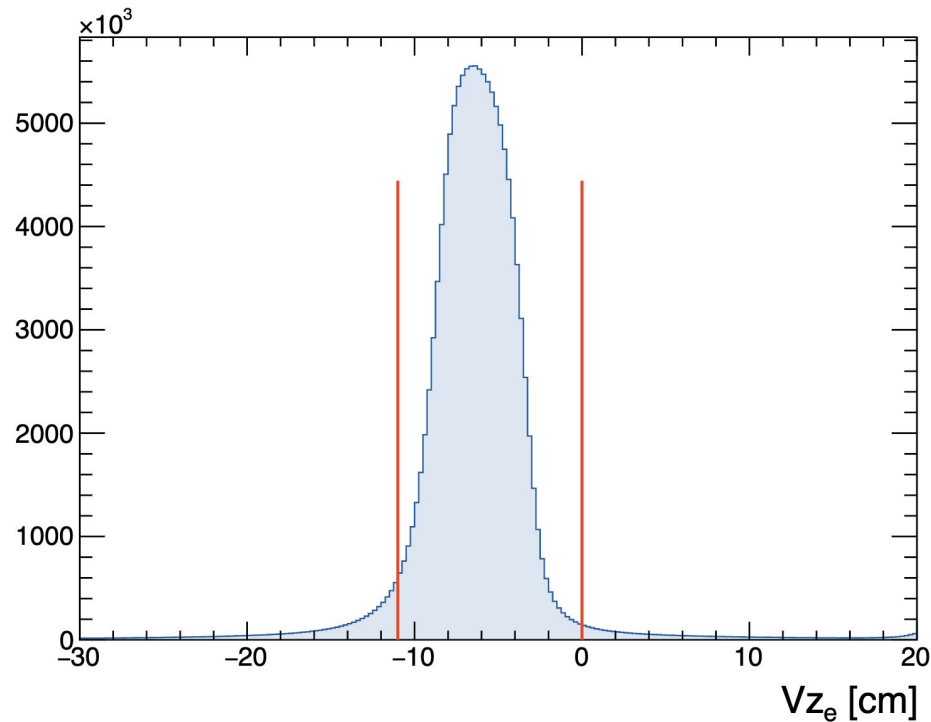
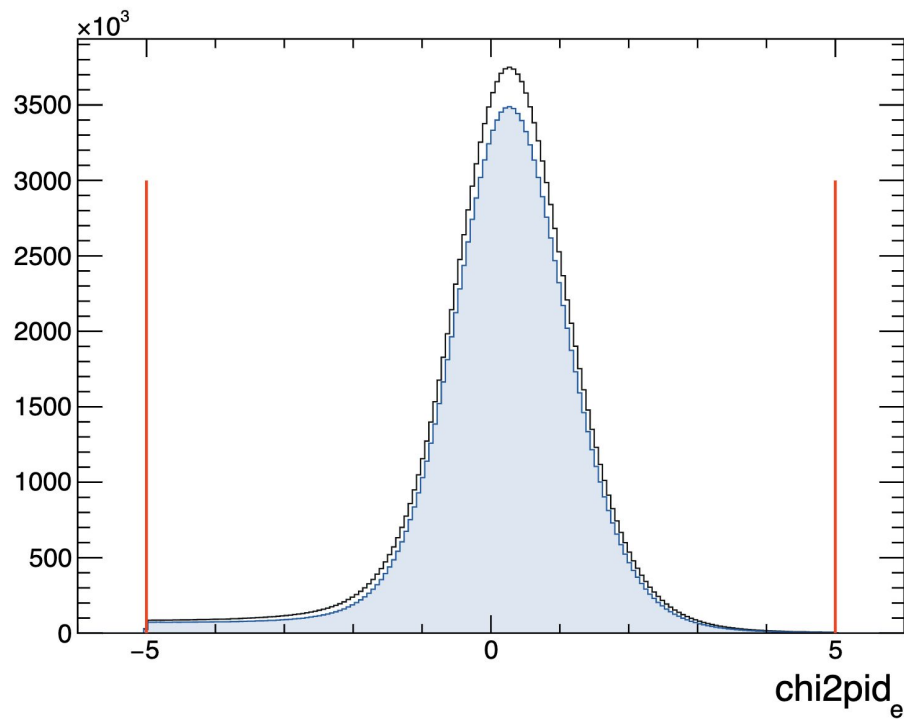
Use all Pass0v7 data, LD2 target.

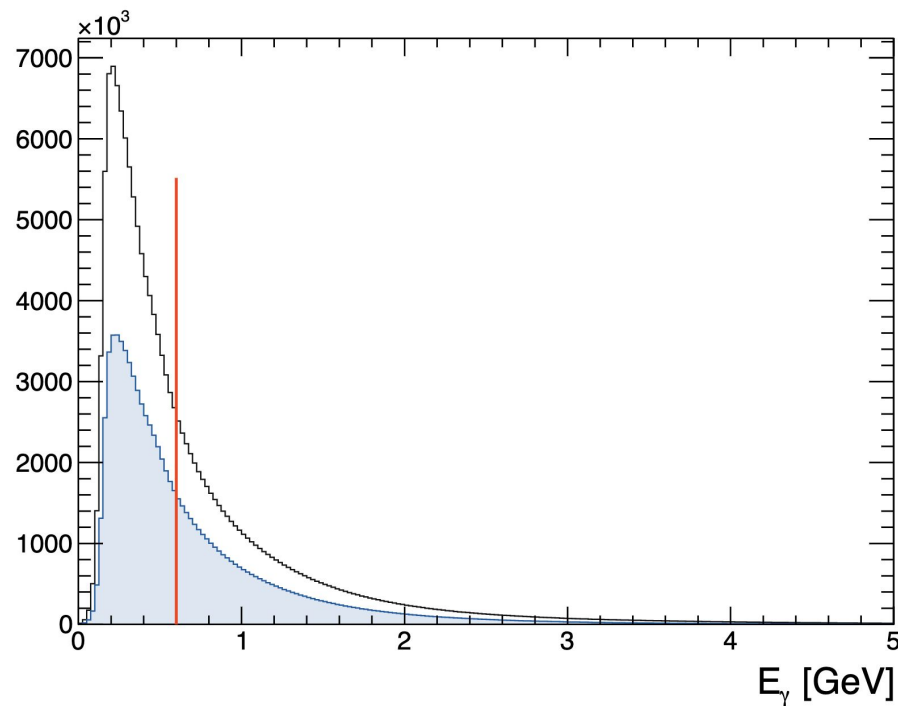
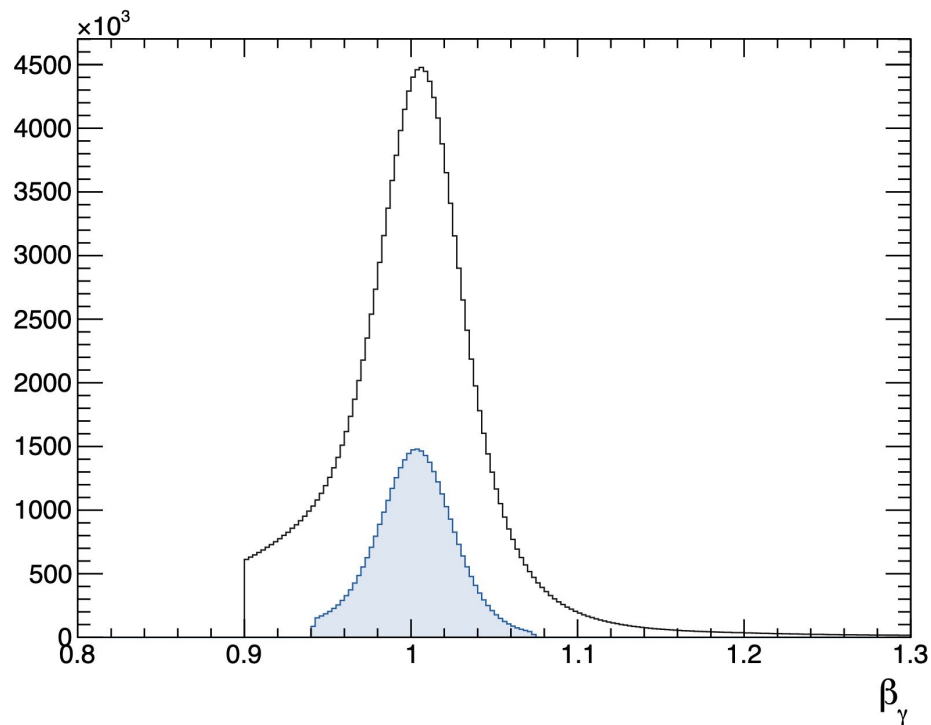
Goal: find the π^0 and estimate the amount.

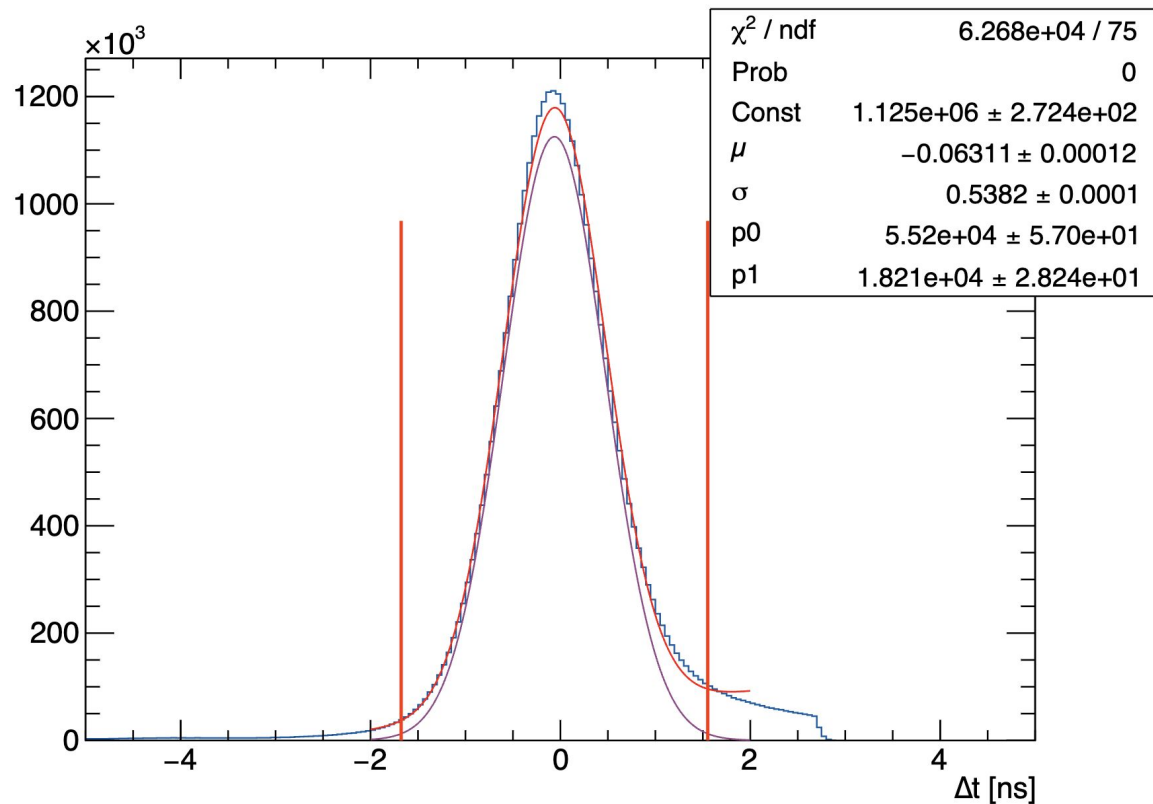
1: ([Analysis note](#) by Stefan Diehl, Andrey Kim, Kyungseon Joo)

- Select trigger electron: pid=11, status<0, $-5 < \chi^2_{\text{pid}} < 5$ and $-11 < V_z < 0$ cm
- Select photon:
 - pid=22 (EB use a cut on $\square > 0.9$ to assign the pid + $E > 0.0001 \text{ GeV}$)
 - $E > 0.6 \text{ GeV} \rightarrow$ reduce radiative photon ¹
 - $-11 < V_z < 0$ cm
 - $\text{sector}_e \neq \text{sector}_\gamma \rightarrow$ reduce radiative photon ¹
 - cut based on the matching of the photon flight time with the path which photon traveled from target to the EC: $-1.68 < t_{\text{EC}} - t_{\text{start}} - L_{\text{EC}} / 30 < 1.55 \text{ ns}$ ([Analysis note](#) by T.Mineeva, W.Brooks, K.Joo)

Radiative photons are dominant at small energies and show a small opening angle in relation to the emitting scattered electron

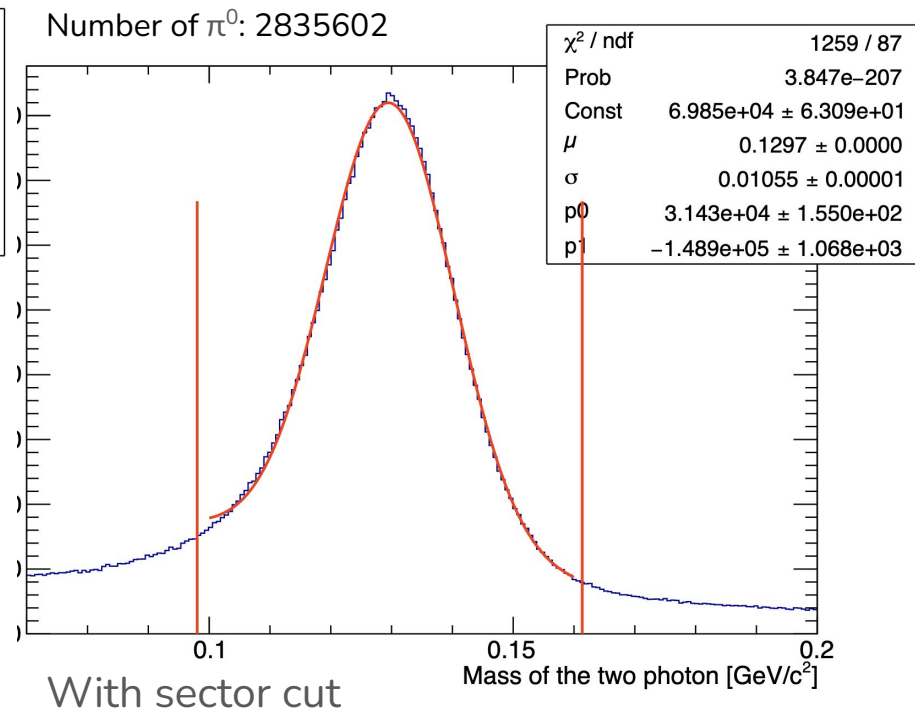
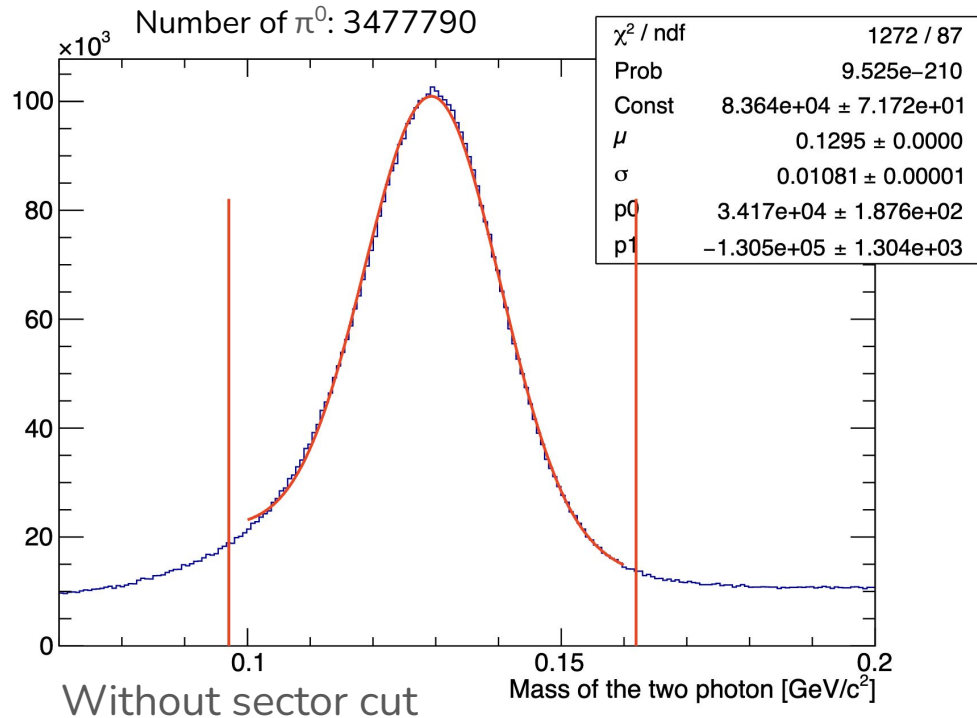






RG-D study: π^0 analysis

Try a cut on the photon sector: sector _{γ_1} = sector _{γ_2} kinematically both photons from neutral pion decay are more likely to end up in the same sector ([Analysis note](#) by Stefan Diehl, Andrey Kim, Kyungseon Joo)



Try a cut on the photon sector: sector _{γ_1} = sector _{γ_2} : kinematically both photons from neutral pion decay are more likely to end up in the same sector ([Analysis note](#) by Stefan Diehl, Andrey Kim, Kyungseon Joo)

Fit the two photons mass distribution with a gaussian and a 1st order polynomial function.

Small improvement with the sector cuts but reduction of the number of π^0 by 20%

Without sector cut

Number of π^0 : 3477790

μ : 0.1295 GeV/c²

σ : 0.01081 GeV/c²

With sector cut

Number of π^0 : 2835602

μ : 0.1297 GeV/c²

σ : 0.01055 GeV/c²

- AI for ALERT:
 - Redo the efficiency/purity analysis using luminosity simulation to have a better background
 - Comparison between conventional and AI track finding
- RG-D analysis:
 - Cross check with Matt the V_x , V_y and V_z mean and sigma for each target
 - Cross check for the nuclear transparency vs Q^2
 - Work on π^0 analysis:
 - Improve π^0 opening angle between the emitting scattered electron and the photon as function of the momentum ([Analysis note](#) by Stefan Diehl, Andrey Kim, Kyungseon Joo)
 - I also would like to work on a AI to identify photon ([Analysis note](#) by Gregory Matousek)
 - Study the E_γ cut effect on the two photons mass